

EEE 310

Communication Systems I Laboratory

Final Project Report

Section: B2 Group: 04

Duplex Communication using LI-FI

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1. Abstract

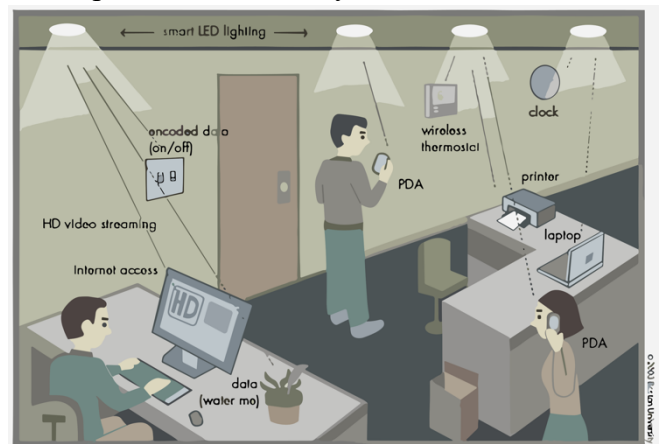
This project presents the development and implementation of a duplex communication system leveraging Light Fidelity (LI-FI) technology. The objective was to create a high-speed, bidirectional communication channel using visible light communication (VLC) principles, specifically utilizing a laser diode and solar panel for transmission and reception, respectively. The system was designed to address the growing demand for wireless communication bandwidth and to offer a secure alternative to radio frequency (RF) systems in sensitive environments. Our findings indicate that such a system is not only viable but also advantageous in terms of data throughput, security, and spectral efficiency. We successfully demonstrated the transmission of binary data encoded in light, received, and decoded accurately over a VLC link. This system's significance lies in its potential application in electromagnetic-sensitive areas like hospitals and aviation, showcasing a scalable solution for future wireless communication needs. The report outlines the design, implementation, and analytical considerations, including public health, safety, and environmental impacts, thereby underscoring the project's innovation within the communication field.

2. Introduction

2.1 Visible Light Communication and Li-Fi

In the contemporary landscape of wireless communication, a significant amount of research is dedicated to enhancing the range and bandwidth of data transmission. With the burgeoning demand for wireless data transfer, the traditional medium of radio waves, despite their global prevalence, is reaching saturation in terms of bandwidth due to limited spectrum availability. This has catalyzed the exploration of Visible Light Communication (VLC) as a promising alternative. VLC offers several advantages, such as minimized noise interference and increased throughput from Mbps to Gbps, establishing it as a superior channel for data transmission [1, 2].

Li-Fi, or Light Fidelity, emerges as a novel technique within the VLC domain, utilizing the optical spectrum to achieve high-speed data transfer. Unlike Wi-Fi, which is optimized for larger spaces, Li-Fi excels in providing secure, high-bandwidth communication in confined areas, mitigating the interference issues associated with radio frequency (RF) waves. The essence of Li-Fi is the transmission of multimedia content through an LED-photodiode pair, creating a communication system that is not only less



Optical Wireless Communication using Li-Fi

complex but also capable of higher data rates and lower power consumption. In our project, we employ Li-Fi technology as the backbone for a duplex communication system. By harnessing LEDs as a source of light, our system demonstrates how a standard communication environment can be revolutionized, offering efficiency and high-speed data transfer without the complexities inherent in acoustic communication systems.

2.2 Li-Fi System with Working Principle & Features

Li-Fi stands as a rapidly evolving field within wireless communications. The conventional structure of a Li-Fi system, as shown in Figure 1, comprises the essential circuit blocks for data transmission, where LEDs function as the transmitting entity, and photosensitive receivers serve as the receiving end.

The core mechanism of Li-Fi is straightforward: binary data, represented by the digits 1 and 0, correspond to the LED's state of being on or off, respectively. This binary encoding is transmitted as light, with the data transfer rate varying depending on the nature of the data and the modulation technique employed. For our duplex communication system, we have implemented On-Off Keying (OOK) modulation due to its simplicity and effectiveness in transmitting binary data through light pulses.

Our system further innovates by incorporating code that enables the LED to transmit user input as binary data, while a solar panel acts as the receiver, capturing the light signals and converting them back into a digital format. This approach not only provides a practical example of VLC but also showcases the potential of Li-Fi in modern communication scenarios, where speed, efficiency, and security are paramount.

3. Design

3.1 Problem Formulation (PO(b))

3.1.1 Identification of Scope

The scope of this project is to design and implement a duplex communication system using LI-FI technology. The capabilities of the system are centered around high-speed, bidirectional data transmission using light as a medium. By utilizing a laser diode for sending data and a solar panel for receiving it, the system aims to establish a proof-of-concept for LI-FI in applications where traditional radio-based wireless communication is impractical or poses security risks.

However, the scope also encompasses certain limitations. The effective range of communication is limited by the intensity and focus of the light source, as well as the sensitivity and placement of the photovoltaic receiver. Moreover, the system requires a line-of-sight between the transmitter and receiver, which is not always feasible in every application. Another constraint is the ambient light interference that can affect the reliability of the data transmission.

3.1.2 Literature Review

Recent advancements in optical communication, particularly LI-FI, suggest it can surpass traditional RF communication in several key areas. LI-FI offers noise minimization and significantly higher throughput, reaching from Mbps to

Gbps, distinguishing it as a superior medium for data transmission [1, 2]. Studies by Srivastava et al. [3] and Ifada et al. [4] have shown practical applications of LI-FI, from highway navigation to user interface designs for system evaluation. Xavier et al. [5] extended LI-FI's functionality by integrating an ultrasonic sensor for triggered data transfer, emphasizing LI-FI's potential for interactive systems.

Despite these advancements, LI-FI technology faces challenges, including the optimization of hardware for low bandwidth applications and addressing noise and errors in data transmission, especially in large-scale data transfers [6,7]. Arduino-based solutions have been a focal point in addressing these challenges due to their flexibility in managing data encoding and decoding [8-10]. The integration of LI-FI into various sectors underscores its growing importance and the necessity for ongoing research to tackle its inherent challenges.

3.1.3 Formulation of Problem

The problem addressed by this project is two-fold. First, the increasing saturation and security concerns associated with RF communication call for an alternative that can operate in sensitive environments without interference. Second, there is a need for a communication system that can deliver high data rates in limited spaces where traditional wireless solutions are not viable.

LI-FI technology presents itself as a solution to these problems. However, its practical implementation faces challenges such as the design of an efficient modulation scheme, minimization of interference from external light sources, and the creation of a system that can seamlessly switch between transmission and reception modes for duplex communication.

3.1.4 Analysis

To address the identified problems, the project employs a LI-FI system utilizing On-Off Keying (OOK) modulation due to its simplicity and low complexity. The choice of OOK allows for a straightforward interpretation of binary states by toggling the laser diode's emission, which directly translates into high or low signals received by the solar panel.

The theoretical background justifies the use of a laser diode for its capability to emit focused and coherent light, which is advantageous for maintaining a strong signal over distance and minimizing dispersion. The solar panel is selected as a receiver due to its ability to detect a wide range of light intensities and convert them into electrical signals, which can then be decoded back into digital data.

The duplex nature of the system is achieved by incorporating a switching mechanism within the code, allowing the microcontroller to toggle between transmitting and receiving modes, thus enabling bidirectional communication. This design approach addresses the scope's limitations by maximizing the potential of the components within the constraints of LI-FI technology.

3.2 Design Method (PO(a))

The design method for our duplex LI-FI communication system is methodical and iterative, focusing on establishing a robust communication link using visible light. The approach is divided into sequential steps, ensuring each component performs optimally before integrating into the larger system. Below is a detailed account of the design methodology using the components listed:

Components List:

1. **Laser Diodes:** High-intensity laser diodes act as transmitters, encoding data into light signals.
2. **Photodiodes:** Sensitive to incoming light, these convert light signals back into electrical signals for data demodulation.
3. **Arduino Uno:** This microcontroller board serves as the central processing unit, controlling the modulation, demodulation, and overall data transmission protocol.
4. **Potentiometer:** Utilized for fine-tuning the circuit parameters, such as adjusting the brightness of the laser diode to optimize transmission.
5. **Solar Panel:** A more extensive light-detecting surface used as an alternative receiver to the photodiodes, extracting data from the laser light by generating voltage variations.
6. **Resistors:** Various resistors are employed to create voltage divider circuits, protecting the components from overvoltage, and ensuring correct voltage levels are applied.

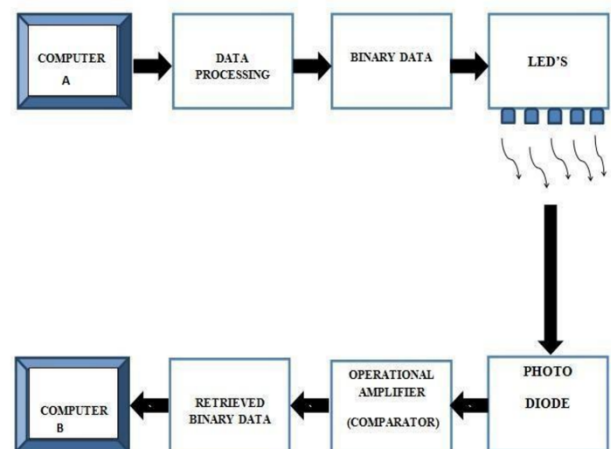
Methodology:

Step 1: Circuit Design The circuit is designed to connect laser diodes to the Arduino's digital pins for data transmission and photodiodes to the analog pins for data reception. A potentiometer is included to adjust the laser diode's intensity, while resistors ensure safe voltage levels for all components.

Step 2: Programming The Arduino is programmed for modulation using On-Off Keying (OOK) for transmission and demodulation for reception. It encodes binary data into light pulses and decodes received light pulses back into binary data.

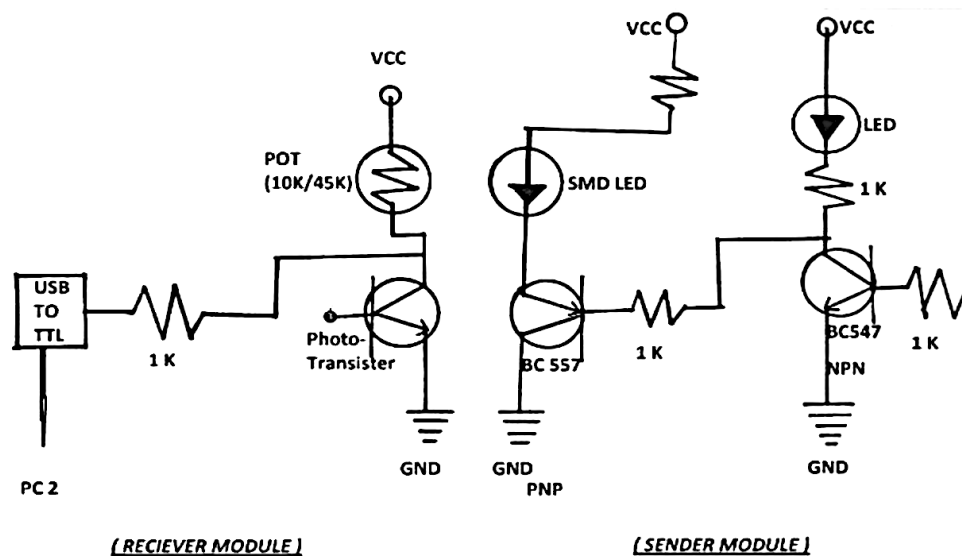
Step 3: Assembly Components are assembled on a breadboard following the circuit design. This setup allows for easy adjustments and troubleshooting during the development process.

Step 4: Testing Testing involves sending and receiving data across the LI-FI system to evaluate performance metrics like data transfer rate and error rate, making necessary adjustments for optimization.

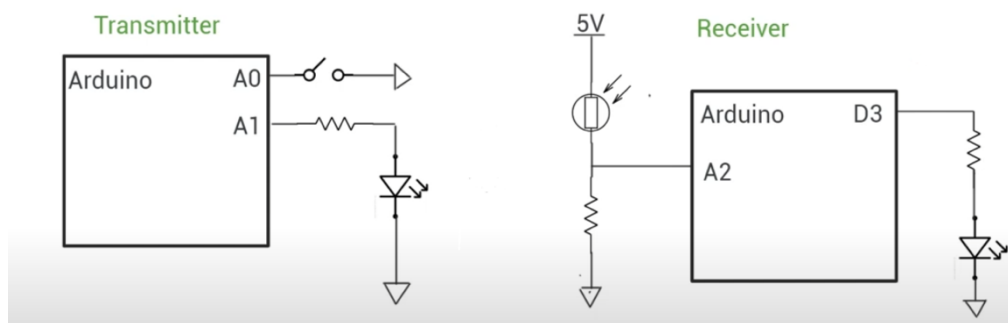


System Block Diagram

3.3 Circuit Diagram



Basic Circuit Diagram

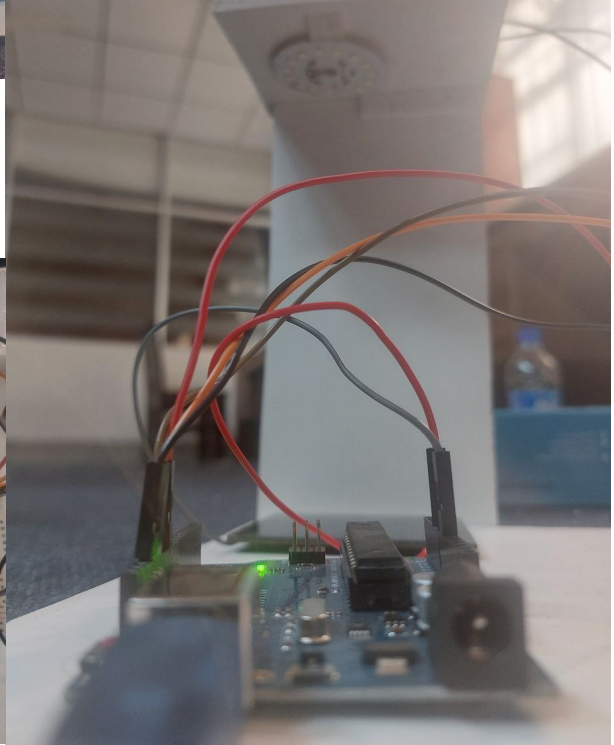
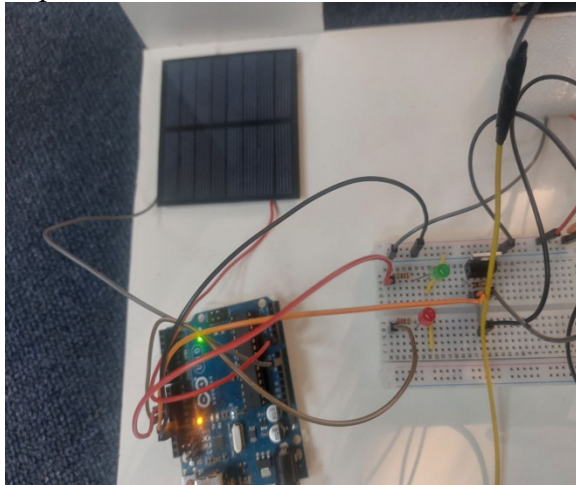
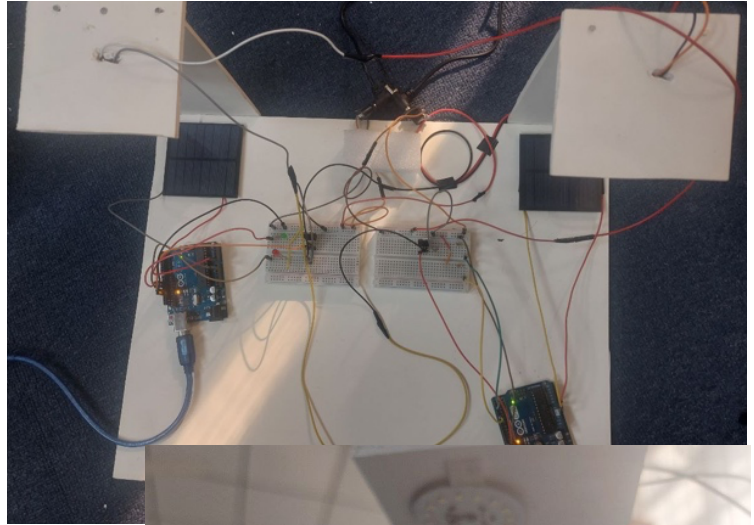


Circuit Diagram with Arduino

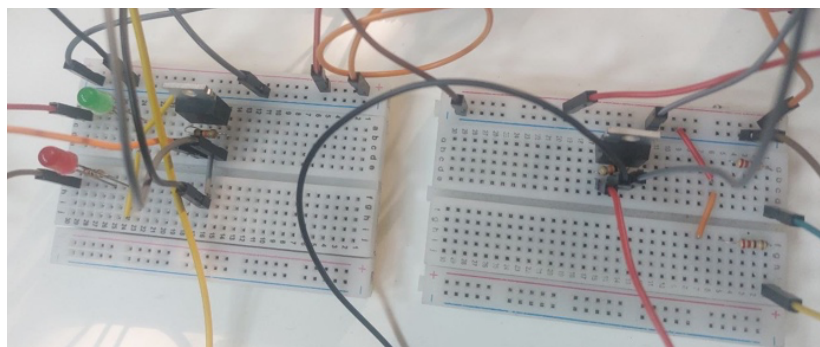
3.4 CAD/Hardware Design

Our LI-FI project's hardware design was conceived to support the Arduino code's requirements for a duplex communication system. Using a breadboard for assembly, we connected laser diodes to the Arduino's digital pins, enabling us to modulate and transmit data as light signals. The solar panel, linked to the Arduino's analog pins, served as the receiver, capturing these signals and converting them back into electrical signals for the Arduino to demodulate.

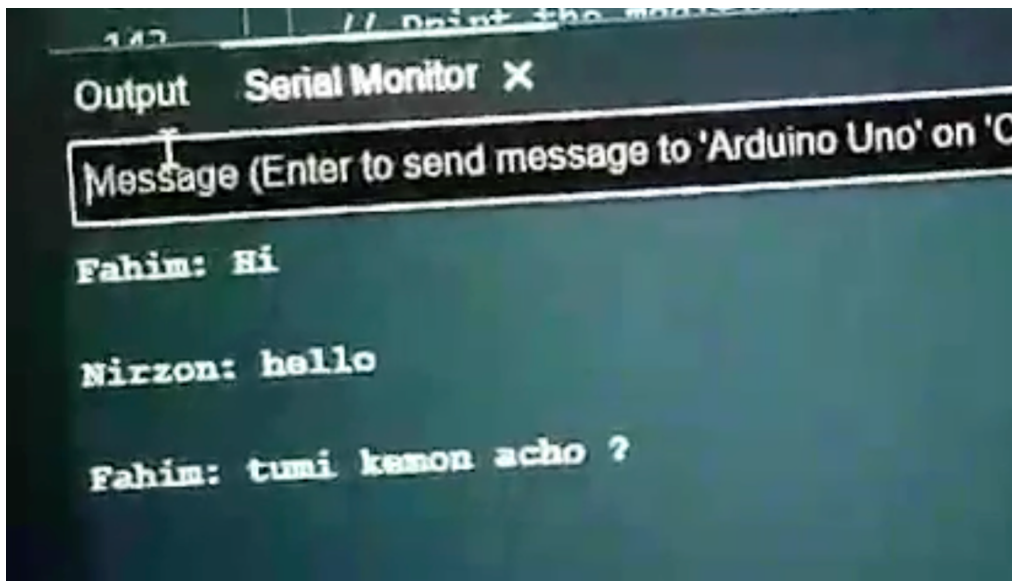
We included potentiometers to adjust the laser diode's output, allowing us to adapt to different distances and light conditions. Resistors were strategically placed to protect the circuit from current surges. The design was compact, modular, and allowed for real-time



adjustments, critical for the iterative testing and development process inherent to our LI-FI system's creation.



Different parts of Hardware Implementations



Message Transferring from one computer to another computer.

3.5 Full Source Code of Firmware

Include the full source code of the firmware used in the project. For example:

```
#define GATE_PIN 13
#define SOLARPIN A0
#define LIGHT_THRESHOLD 500 // Adjust this value
                             based on your environment
#define ARRAY_SIZE 500      // Size of the array to
                             hold the bits
#define GREEN_LED_PIN 10    // Pin for the green LED
#define YELLOW_LED_PIN 11  // Pin for the yellow
                             LED

char bitsArray[ARRAY_SIZE]; // Array to hold the bits
int currentIndex = 0;        // Index to keep track
                             of the current position in the array
bool istransmit = true;

void setup() {
    Serial.begin(9600);
    pinMode(GATE_PIN, OUTPUT); // Set laser control
    pin as output
    digitalWrite(GATE_PIN, HIGH);
    pinMode(GREEN_LED_PIN, OUTPUT); // Set green LED
    pin as output
    pinMode(YELLOW_LED_PIN, OUTPUT); // Set yellow LED
    pin as output
}
```

```

void loop() {
    if (!istransmit) {
        transmit();
    } else {
        receive();
    }
    updateLEDs(); // Update the LED states based on
the current mode
}

void updateLEDs() {
    if (istransmit) {
        digitalWrite(GREEN_LED_PIN, HIGH); // Turn on the
green LED for transmit mode
        digitalWrite(YELLOW_LED_PIN, LOW); // Turn off
the yellow LED
    } else {
        digitalWrite(GREEN_LED_PIN, LOW); // Turn off
the green LED
        digitalWrite(YELLOW_LED_PIN, HIGH); // Turn on
the yellow LED for receive mode
    }
}

void transmit() {
    // Clear previously stored information
    String userInput = "";

    // Wait for user input in the Serial Monitor
    while (Serial.available() > 0) {
        char c = Serial.read();
        if (c == '\n' || c == '\r') {
            // Process the user input
            digitalWrite(GATE_PIN, LOW); // Start
Indicator
            delay(100);
            digitalWrite(GATE_PIN, HIGH);
            //delay(1000);
            processAndTransmit(userInput);
            // Clear the stored input for the next iteration
            userInput = "";
            // Set istransmit to true to switch to receiving
mode
            istransmit = true;
        } else {
            // Append the character to the current user
input
            userInput += c;
        }
    }
}

```

```

    }

    // Add a small delay for stability
    delay(100);
}

void processAndTransmit(String inputText) {
    // Display the user input
    Serial.print("Nirzon: ");
    Serial.println(inputText);

    // Display the corresponding binary value
    //Serial.print("Binary Value: ");
    int i = 0;
    for (i; i < inputText.length(); i++) {
        int binaryValue = inputText[i];

        // Print the binary representation
        for (int j = 7; j >= 0; j--) {
            //Serial.print((binaryValue >> j) & 1);

            // Transmit the bit using the laser module
            digitalWrite(GATE_PIN, !((binaryValue >> j) &
1));
            delay(100); // Adjust delay as needed
        }

        //Serial.print(" "); // Add space between
characters
    }
    // delay(1000);

    if (i == inputText.length()) {
        digitalWrite(GATE_PIN, HIGH);
    }

    Serial.println(); // Print a newline for
readability
}

void receive() {
    int consecutiveZeroCount = 0; // Track consecutive
zero bits

    bool printingEnabled = false; // Flag to enable
printing after the first light hits the solar panel

    while (consecutiveZeroCount < 8 && currentIndex <
ARRAY_SIZE) {
        int sensorValue = analogRead(SOLARPIN);

```

```

        if (!printingEnabled && sensorValue >
LIGHT_THRESHOLD) {
            printingEnabled = true; // Enable printing once
the light hits the solar panel
        }

        //delay(100);

        if (printingEnabled) {
            if (sensorValue > LIGHT_THRESHOLD) {
                bitsArray[currentIndex++] = '1'; // Store '1'
in the array
                consecutiveZeroCount = 0; // Reset
consecutive zero count
            } else {
                bitsArray[currentIndex++] = '0'; // Store '0'
in the array
                consecutiveZeroCount++; // Increment
consecutive zero count
            }
        }

        delay(100);
    }
    // Null-terminate the array
    bitsArray[currentIndex] = '\0';

    // Store bits into another array without the first
bit
    char newBitsArray[ARRAY_SIZE];
    for (int i = 0; i < currentIndex; i++) {
        newBitsArray[i] = bitsArray[i + 1]; // Start
copying from the second element of bitsArray
    }

    // Null-terminate the new array
    newBitsArray[currentIndex - 1] = '\0';

    // Print the new array
    //Serial.println("New bits stored in the array
without the first bit:");
    //Serial.println(newBitsArray);

    // Calculate the length of newBitsArray
    int n = strlen(newBitsArray) % 8;

    if (n == 0) {

```

```

    newBitsArray[strlen(newBitsArray) - 8] = '\0';
} else {

    // Remove the last n indices from newBitsArray
    int newLength = strlen(newBitsArray) - n;
    newBitsArray[newLength] = '\0';
}
// Print the modified array
//Serial.println("Modified newBitsArray (removed
the last n indices:");
//Serial.println(newBitsArray);
// Convert every 8 bits into corresponding ASCII
and store them in a string
String asciiString = "";
for (int i = 0; i < strlen(newBitsArray); i += 8) {
    // Extract 8 bits substring
    char substring[9];
    strncpy(substring, newBitsArray + i, 8);
    substring[8] = '\0';

    // Convert substring to integer
    int intValue = strtol(substring, NULL, 2);

    // Convert integer to ASCII and append to the
string
    asciiString += char(intValue);
}

// Print the ASCII string
Serial.print("Nirzon: ");
Serial.println(asciiString);
Serial.println();

// Empty the array
for (int i = 0; i < ARRAY_SIZE; i++) {
    bitsArray[i] = '\0';
}
currentIndex = 0; // Reset currentIndex to 0 for
the next iteration
istransmit = false;
}

```

4. Implementation

4.1 Description

The implementation process involved constructing the LI-FI communication

system on a breadboard, enabling easy modification and troubleshooting. Initial challenges included minimizing interference from ambient light and ensuring a clear line-of-sight for the laser diode and the photodiode/solar panel receiver. These were overcome by testing in controlled light conditions and using focused light sources and sensitive receivers.

4.2 Experiment and Data Collection

The experimental setup comprised a laser diode connected to the Arduino for signal transmission and a solar panel/photodiode connected to another Arduino for signal reception. Data collection was conducted by transmitting a set of predefined binary sequences and recording the received sequences for comparison.

4.3 Data Analysis

Data was analyzed by comparing the transmitted binary sequences against the received ones to identify any discrepancies. The bit error rate was calculated to assess the performance, and throughput was measured to determine the system's data transfer rate.

4.4 Results

The LI-FI system successfully transmitted and received binary data sequences with a low bit error rate, demonstrating the system's potential for high-speed data transfer. Observations also indicated a stable performance in controlled lighting conditions.

5. Design Analysis and Evaluation

5.1 Novelty

Our LI-FI project introduces several novel aspects to the field of LI-FI technology. By utilizing a solar panel as the receiver, the project explores alternative, potentially more sensitive light detection methods. This innovation could lead to LI-FI systems that are more efficient in ambient light conditions, broadening the technology's applicability. Additionally, the use of a simple, yet effective, modulation technique (OOK) and off-the-shelf components lowers the barrier to entry for LI-FI experimentation and could spur further grassroots innovation in the field.

5.2 Design Considerations (PO(c))

5.2.1 Considerations to Public Health and Safety

The system is designed with low-intensity diodes, which pose no threat to public health and safety under normal operation. The light emissions are within safe exposure levels, and since the communication is optical, it avoids the potential health concerns associated with long-term exposure to RF waves.

5.2.2 Considerations to the Environment

The environmental impact of the LI-FI system is minimal. The components used consume low amounts of electricity, and the system does not emit any hazardous substances or RF radiation. It represents an eco-friendly alternative to traditional wireless communication methods.

5.2.3 Considerations to Cultural and Societal Needs

The system provides a solution for secure communication in environments where RF signals are inappropriate or banned, such as hospitals and schools, aligning with societal needs for privacy and security.

5.3 Limitations of Tools (PO(e))

The primary limitation of the tools used, such as the breadboard and simple electronic components, is scalability. While suitable for a proof-of-concept, they may not meet the demands of a commercial-grade system.

5.4 Impact Assessment (PO(f))

5.4.1 Assessment of Societal and Cultural Issues

The project's implementation reflects a growing cultural emphasis on data security and the societal trend towards sustainable technology.

5.4.2 Assessment of Health and Safety Issues

The design adheres to health and safety standards by using components that are safe for public interaction and do not require specialized handling.

5.4.3 Assessment of Legal Issues

The use of LI-FI technology currently faces few legal restrictions, but the project considers potential future regulations regarding optical transmission.

5.5 Sustainability and Environmental Impact Evaluation (PO(g))

Long-term environmental impacts are expected to be positive, as LI-FI systems like ours could reduce dependence on RF communication, leading to energy savings and reduced electronic waste.

5.6 Ethical Issues (PO(h))

Ethically, the project promotes inclusivity by providing an alternative communication method that can operate alongside existing technologies, ensuring broader access to information without contributing to electronic pollution or RF spectrum overcrowding.

6. Reflection on Teamwork (PO(i))

6.1 Mode of Teamwork

Our team collaborated efficiently through a combination of scheduled weekly meetings, shared online workspaces for document collaboration, and task delegation based on individual skills and interests. All members were involved

in major decision-making processes and hands-on tasks such as the assembly of the light board and the voltage and current sensing circuit.

6.2 Diversity Statement of Team

The diversity within our team, encompassing different backgrounds, areas of expertise, and problem-solving approaches, significantly enriched the project. This amalgamation of perspectives fostered a creative environment that enhanced the project's innovation and allowed for a robust exchange of ideas.

6.3 Timeline of Project Implementation

Our team Consists of 6 members ID 2006126, ID 2006127, ID 2006128 and ID 2006130

Here's the list of our contributions:

Task Accomplished	Date	Member(s) Involved
Project Proposal (Doc)		All Members
Project Proposal (Presentation)		2006126
Purchased components	03/02/24	All Members
Built a basic Li-Fi with laser light	15/02/24	All Members
Assembled the main circuit	24/02/24	All Members
Displayed the accurate voltage and current values on lcd	01/03/24	All Members

7. Cost Analysis (PO(k))

- Arduino Uno (2 pieces): BDT 1,050 x 2 = BDT 2,100
 - LED strips: BDT 500 (estimated)
 - Mosfets: BDT 100
 - Resistors: BDT 50
 - Breadboard (2 pieces): BDT 200 x 2 = BDT 400
 - Solar panel (2 pieces, small): BDT 1,000
 - Potentiometer: BDT 50
 - Jumper wires: BDT 100
- Total: BDT 4,400

8. Future Work (PO(l))

The LI-FI technology demonstrated in our project opens several avenues for future research and development. Potential future improvements to our duplex communication system using LI-FI may include:

1. **Increased Range and Efficiency:** Research into advanced modulation techniques and more sensitive receivers could extend the range and improve the efficiency of LI-FI systems.

2. **Integration with IoT Devices:** Embedding LI-FI capability into Internet of Things (IoT) devices could enable faster M2M communication and facilitate smart home applications.
3. **Data Transmission Security:** Further work on the encryption methods used in LI-FI could enhance the security of data transmission, making it more robust against eavesdropping.
4. **Interference Mitigation:** Studying the impact of ambient light and devising methods to mitigate its effects would be crucial for the reliability of LI-FI in various lighting conditions.
5. **Seamless Handover:** Ensuring seamless handover between LI-FI and other technologies such as Wi-Fi and LTE for uninterrupted connectivity as users move through different environments.
6. **Power Consumption Optimization:** Exploring ways to lower the power consumption of LI-FI systems for their viability in battery-powered devices.
7. **Application in Congested Environments:** Implementing LI-FI in high-density settings like conferences and concerts where traditional wireless systems struggle due to congestion.
8. **Broadband Internet Provisioning:** Utilizing LI-FI to provide broadband internet in areas where RF communications are restricted, such as hospitals and airplanes.
9. **Enhanced Data Rates:** Leveraging emerging technologies to increase the data rates achievable with LI-FI, making it competitive with or superior to current wireless standards.
10. **Underwater Communication:** Adapting the system for underwater communication where RF signals cannot penetrate effectively.

The project's future iterations will focus on addressing these aspects, with an emphasis on practical implementations and real-world testing to bring LI-FI closer to widespread adoption.

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